

# Ashutosh Jha

Braunschweig, Germany

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PhD Student, Machine Learning

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[Google Scholar](#)

Languages: English (C1), Deutsch (B1)

## Education

PhD in Machine Learning

Sep 2026 - Present

*Institute of Artificial Intelligence, TU Braunschweig, Germany*

Advisor: Prof. Dr. Michel Besserve

*Causal and probabilistic representation learning for complex systems.*

M.Sc. Quantitative Data Science Methods, Machine Learning & Econometrics

Oct 2023 - Jun 2026

*Eberhard Karls Universität Tübingen, Germany*

Grade: 1.48/5.0 (1.0 = best)

*Thesis: Optimal Transport in linear Independent Component Analysis, at the Max Planck Institute for Intelligent Systems.*

Advisors: Dr. Simon Buchholz, Prof. Dr. Michel Besserve, Prof. Dr. Joachim Grammig

Erasmus Exchange Semester — Financial Economics and Machine Learning

Feb 2025 - Jun 2025

*Katholieke Universiteit Leuven, Belgium*

Grade: 1.5/5.0 (1.0 = best)

B.E. Electrical and Electronics Engineering

Aug 2016 - May 2020

*Birla Institute of Technology and Science, Pilani, Goa Campus, India*

Grade: 8.12/10

## Publications

### Linear Independent Component Analysis via Optimal Transport

Ashutosh Jha, Michel Besserve, Simon Buchholz.

*9th Workshop on Tractable Probabilistic Modeling (TPM), UAI 2026, Amsterdam.*

[arXiv:2607.14081](#) | [OpenReview](#) | [PDF](#)

- Measures non-Gaussianity by the squared Wasserstein distance to a standard Gaussian rather than by proxy contrast functions, with a proof that this objective is maximised exactly when a projection recovers an independent component. The resulting OT-ICA algorithm outperforms proxy-based methods on simulated data and is validated on EEG artifact removal and econometric price discovery.

## Research Experience

### Student Research Assistant — Perceiving Systems

Nov 2025 - Jun 2026; Jul 2024 - Jun 2025

at [Max Planck Institute for Intelligent Systems](#), Tübingen, Germany

- Extended the NICE toolbox to process non-verbal communication data by integrating computer-vision detectors for pose estimation, emotion detection, and head orientation. Tools: Python, PyTorch.
- Containerised the application and managed CI/CD pipelines to make experimental models reproducible across machines. Tools: Docker, Apptainer, GitLab CI/CD, Hugging Face, Linux.

### Quantitative Analyst Intern — Financial Stability

Jul 2025 - Sep 2025

at [Deutsche Bundesbank](#), Frankfurt, Germany

- Modelled investment-fund portfolio vulnerabilities against market, monetary-policy, and geopolitical shocks by stress-testing econometric, probabilistic decision-tree, and foundation tabular (TabPFN) time-series models. Tools: Python, SQL, Git.

## Industry Experience

### Software Engineer, HSBC Holdings PLC, Pune, India

Sep 2020 - Sep 2023

Cloud economics and FinOps: statistical cost-benefit models (\$500K savings, 1.5× vendor discounts) and real-time ETL pipelines at 4× lower latency. Python, SQL, GCP, AWS, Jenkins.

### Software Engineering Intern, Nvidia Corporation, Bengaluru, India

Aug 2019 - Dec 2019

Real-time visualisation tool for root-cause analysis of SoC CPU boot-sequence verification. Python, Linux, shell scripting.

## Selected Projects

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**TuebiFit — Full-Stack Agentic Application** (Heilbronn Hackathon 2026) *Mar 2026*  
[Git Repository](#) — LLM agents and Model Context Protocol (MCP) servers orchestrated with LangChain and LangGraph, with real-time pose estimation (MediaPipe), deployed on Google Cloud Run.

**Probabilistic Asset Pricing**, Master Seminar, Universität Tübingen *Nov 2024 - Jan 2025*  
[Git Repository](#) — Bayesian asset pricing using asset-pricing theory as a prior and a state-space representation (Kalman filters) to estimate posterior prices, anomalies, and risk premia.

## Awards

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**Best Pitch & Co-creation Prize**, Neckarthon, Startup Center Tübingen *Nov 2024*  
Dual prize for “Tue-bi-smart”, an ML-enabled transport prototype for Stadtwerke Tübingen.

**Prime Minister’s Scholarship Scheme**, Government of India *2017*  
Awarded for academic excellence during undergraduate studies.

## Technical Skills

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**Machine Learning & Research:** *PyTorch, deep learning, probabilistic & Bayesian modelling, optimal transport, causal inference, time-series forecasting, LLM agents (LangChain, LangGraph), Hugging Face, TabPFN.*

**Programming & Databases:** *Python, SQL, R, MATLAB, VBA.*

**MLOps & Tools:** *Git, Docker, Apptainer, CI/CD (GitLab CI/CD, Jenkins), HTCondor, Linux, LaTeX.*

**Cloud & BI:** *GCP (Compute, BigQuery, Cloud Run), AWS (Lambda, S3), Power BI, MS Excel.*

**Relevant Coursework:** *Machine Learning, Deep Learning, Probabilistic Machine Learning, Optimization, Bayesian Modelling, Advanced Time Series Analysis, Machine Learning in Econometrics, Financial Economics.*